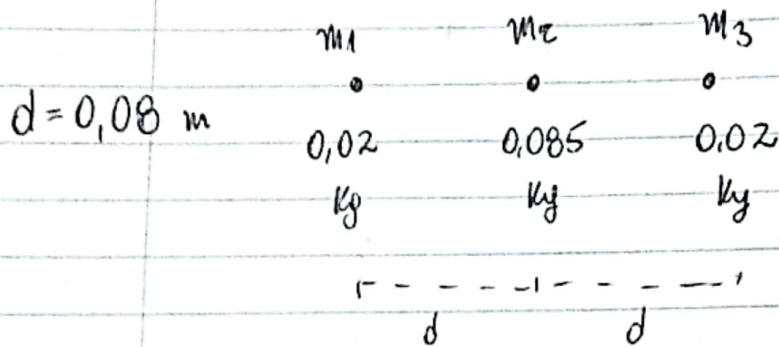


GOUA 4

(4)

$$q = 2 \mu C$$



$$U = K \left[\frac{q_1 q_2}{d} + \frac{q_2 q_3}{d} + \frac{q_1 q_3}{2d} \right]$$

$$U = \frac{Kq^2}{d} \left[1 + 1 + \frac{1}{2} \right]$$

$$= 1,125 \text{ [J]}$$

$$K + U = \text{cte}$$

$$\text{si } r \rightarrow \infty \quad U \rightarrow 0$$

$$\cancel{K_i + U_i} = \cancel{K_F + U_F}$$

$$U_i = K_F$$

$$\rightarrow U = 1,125 \text{ [J]}$$

$$\frac{1}{2} m_1 v_1^2 = 1,125$$

$$v_1 = 10,60 \text{ m/s}$$

$$\frac{1}{2} m_3 v_3^2 = 1,125$$

$$v_3 = 10,60 \text{ m/s}$$

$$\frac{1}{2} m_2 v_2^2 = 1,125$$

$$v_2 = 5,14 \text{ m/s}$$